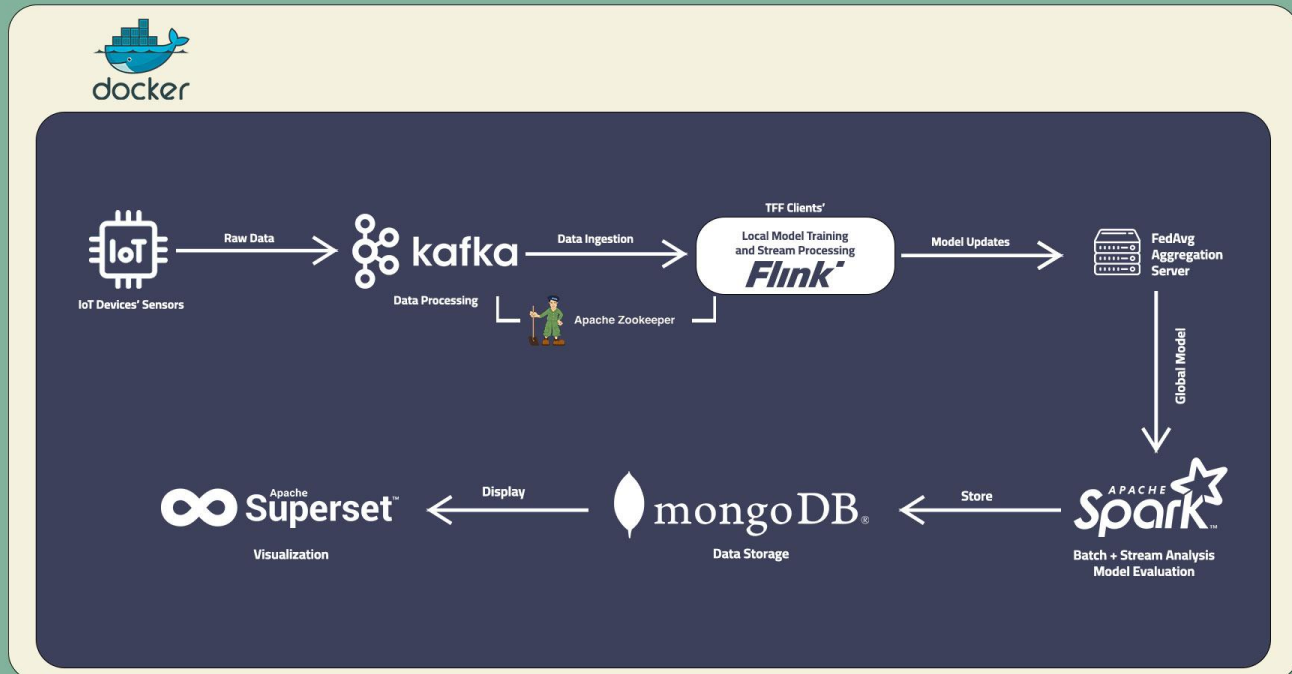


FLEAD: Federated Deep Learning for IoT-Based Anomaly Detection in Real-Time Monitoring



Description: The architecture is a federated deep learning system for real-time anomaly detection in IoT environments, deployed with Docker for scalability. Raw data from Sensors Send to Apache Kafka for ingestion and processing, managed by Zookeeper. Apache Flink clients perform local model training and stream processing, sending model updates to a FedAvg aggregation server that combines them into a global model. Apache Spark then conducts batch and stream analytics for anomaly detection and model evaluation. Processed data is stored in MongoDB and visualized through Apache Superset dashboards, and scalable IoT monitoring.

Work Plan:

Member 1 – Data Ingestion & Preprocessing - SU YOUNG and ROBERT

Member 2 – Local Training (Federated Learning) - IMED and AMIR

Member 3 – Federated Aggregation & Ensemble - IMED and AMIR

Member 4 – Batch Analytics & Storage - SU YOUNG and ROBERT

Member 5 – Visualization & Deployment - YUSIF and AMIR